



Programming Fundamentals — Assignment

Lab 04: Loops in C++

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Github Repo: <https://github.com/aaliyann-exe/programming-fundamentals-lab>

Task 1: A student is reading a 200-page book and reads 40 pages every day. Using a while loop, display the total pages read at the end of each day and finally print how many days it took to finish the book.

Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      const int TOTAL_PAGES = 200;
7      const int READ_SPEED = 40;
8      int pagesRead = 0, day = 0;
9
10     while (pagesRead < TOTAL_PAGES) {
11
12         pagesRead += READ_SPEED;
13         day++;
14         cout << "Day " << day << ": " << pagesRead << " pages read" << endl;
15
16     }
17
18     cout << "Book finished in " << day << " days" << endl;
19
20     return 0;
21
22 }
```

Output:

```
Day 1: 40 pages read
Day 2: 80 pages read
Day 3: 120 pages read
Day 4: 160 pages read
Day 5: 200 pages read
Book finished in 5 days
```

Task 2: Registration form. (a) Ask for the user's age and keep asking until it is between 18 and 60. (b) Then ask the user to create a 4-digit PIN (1000 to 9999) and keep asking until it is valid. (c) Finally ask the user to re-enter the PIN to confirm; allow only 3 attempts and display "Account created" or "Registration failed".

Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int age, pin, confirmPin;
7      int attempts = 1;
8
9      // (a) Age
10     cout << "Enter your age: ";
11     cin >> age;
12     while (age < 18 || age > 60) {
13         cout << "Invalid! Age must be between 18 and 60: ";
14         cin >> age;
15     }
16
17     // (b) PIN
18     do {
19         cout << "Create a 4-digit PIN: ";
20         cin >> pin;
21         if (pin < 1000 || pin > 9999)
22             cout << "Invalid PIN! It must be between 1000 and 9999." << endl;
23     } while (pin < 1000 || pin > 9999);
24
25     // (c) confirm PIN
26     cout << "Re-enter PIN to confirm: ";
27     cin >> confirmPin;
28     while (confirmPin != pin && attempts < 3) {
29         cout << "PIN does not match. Try again: ";
30         cin >> confirmPin;
31         attempts++;
32     }
33
34     if (confirmPin == pin)
35         cout << "Account created" << endl;
36     else
37         cout << "Registration failed" << endl;
38
39     return 0;
40 }
41 }
```

Output:

```
Enter your age: 8
Invalid! Age must be between 18 and 60: 20
Create a 4-digit PIN: 333333
Invalid PIN! It must be between 1000 and 9999.
Create a 4-digit PIN: 6767
Re-enter PIN to confirm: 33
PIN does not match. Try again: 6767
Account created
```

Task 3: Write a menu-driven mobile top-up program using a do-while loop. Menu: 1. Rs. 100 load 2. Rs. 200 load 3. Rs. 500 load 4. Show total and Exit. The menu repeats after every choice, an invalid choice shows an error message, and choosing 4 prints how many recharges were done and the total amount.

Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int choice;
7      int recharges = 0, total = 0;
8
9      do {
10
11          cout << "\n--- Mobile Top-Up ---" << endl;
12          cout << "1. Rs. 100 load\n2. Rs. 200 load\n3. Rs. 500 load\n4. Show Total and Exit" << endl;
13          cout << "Choice: ";
14          cin >> choice;
15
16          switch (choice) {
17
18              case 1:
19                  total += 100;
20                  recharges++;
21                  cout << "Rs. 100 loaded." << endl;
22                  break;
23
24              case 2:
25                  total += 200;
26                  recharges++;
27                  cout << "Rs. 200 loaded." << endl;
28                  break;
29
30              case 3:
31                  total += 500;
32                  recharges++;
33                  cout << "Rs. 500 loaded." << endl;
34                  break;
35
36              case 4:
37                  break;
38
39              default:
40                  cout << "Invalid choice." << endl;
41
42          }
43      } while (choice != 4);
44
45      cout << "\nRecharges done: " << recharges << endl;
46      cout << "Total amount: Rs. " << total << endl;
47
48      return 0;
49  }
```

Output:

```
--- Mobile Top-Up ---
1. Rs. 100 load
2. Rs. 200 load
3. Rs. 500 load
4. Show Total and Exit
Choice: 3
Rs. 500 loaded.

--- Mobile Top-Up ---
1. Rs. 100 load
2. Rs. 200 load
3. Rs. 500 load
4. Show Total and Exit
Choice: 4

Recharges done: 1
Total amount: Rs. 500
```

Task 4: Using a for loop, take a number N from the user and print a table of numbers from 1 to N with their squares and cubes (use \t). At the end also print the sum of all the cubes.

Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int n, sumCubes = 0;
7
8      cout << "Enter N: ";
9      cin >> n;
10
11     cout << "Number\tSquare\tCube" << endl;
12
13     for (int i = 1; i <= n; i++) {
14
15         cout << i << "\t" << i * i << "\t" << i * i * i << endl;
16         sumCubes += i * i * i;
17
18     }
19
20     cout << "Sum of cubes = " << sumCubes << endl;
21
22     return 0;
23
24 }
```

Output:

```
Enter N: 6
Number  Square  Cube
1       1      1
2       4      8
3       9     27
4      16     64
5      25    125
6      36    216
Sum of cubes = 441
```

Task 5: Write a program that asks how many students are in a class, then reads each student's marks (0–100, validated using a loop) using a for loop. Using counters, display how many students got grade A (80 or above), how many passed (50 or above), and how many failed.

Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      int students, marks;
7      int gradeA = 0, passed = 0, failed = 0;
8
9      cout << "How many students are in the class? ";
10     cin >> students;
11
12     while (students < 1) {
13         cout << "Invalid! Enter at least 1 student: ";
14         cin >> students;
15     }
16
17     for (int i = 1; i <= students; i++) {
18
19         cout << "Enter marks of student " << i << ": ";
20         cin >> marks;
21         while (marks < 0 || marks > 100) {
22             cout << "Invalid! Marks must be between 0 and 100: ";
23             cin >> marks;
24         }
25
26         if (marks >= 80)
27             gradeA++;
28
29         if (marks >= 50)
30             passed++;
31         else
32             failed++;
33     }
34
35     cout << "Grade A: " << gradeA << endl;
36     cout << "Passed: " << passed << endl;
37     cout << "Failed: " << failed << endl;
38
39     return 0;
40
41 }
42 }
```

Output:

```
How many students are in the class? 0
Invalid! Enter at least 1 student: 5
Enter marks of student 1: 20
Enter marks of student 2: 67
Enter marks of student 3: 82
Enter marks of student 4: 42
Enter marks of student 5: 3
Grade A: 1
Passed: 2
Failed: 3
```

Task 6: Write a cashier program that reads item prices until the user enters 0 (sentinel). Display the number of items, the total, the highest price and the average price. If the total is more than Rs. 1000, give a 10% discount and display the final payable amount. If no items are entered, display "No items entered."

Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      const int SENTINEL = 0;
7      int price, items = 0, total = 0, highest = 0;
8
9      cout << "Enter item price (0 to stop): ";
10     cin >> price;
11
12     while (price != SENTINEL) {
13
14         total += price;
15         items++;
16         if (price > highest)
17             highest = price;
18
19         cout << "Enter item price (0 to stop): ";
20         cin >> price;
21     }
22
23     if (items > 0) {
24
25         float average = (float)total / items;
26         float payable = total;
27
28         cout << "Number of items: " << items << endl;
29         cout << "Total: Rs. " << total << endl;
30         cout << "Highest price: Rs. " << highest << endl;
31         cout << "Average price: Rs. " << average << endl;
32
33         if (total > 1000) {
34             float discount = total * 0.10;
35             payable = total - discount;
36             cout << "Discount (10%): Rs. " << discount << endl;
37         }
38
39         cout << "Payable amount: Rs. " << payable << endl;
40     } else {
41         cout << "No items entered." << endl;
42     }
43
44     return 0;
45 }
46
47
48 }
```

Output:

```
Enter item price (0 to stop): 40
Enter item price (0 to stop): 60
Enter item price (0 to stop): 20
Enter item price (0 to stop): 0
Number of items: 3
Total: Rs. 120
Highest price: Rs. 60
Average price: Rs. 40
Payable amount: Rs. 120
```

Pseudocode:

START

SENTINEL = 0, items = 0, total = 0, highest = 0

INPUT price

WHILE price != SENTINEL

total = total + price

items = items + 1

IF price > highest THEN

highest = price

END IF

INPUT price

END WHILE

IF items > 0 THEN

average = total / items

payable = total

PRINT "Number of items: ", items

PRINT "Total: Rs. ", total

PRINT "Highest price: Rs. ", highest

PRINT "Average price: Rs. ", average

IF total > 1000 THEN

discount = total * 0.10

payable = total - discount

PRINT "Discount (10%): Rs. ", discount

END IF

PRINT "Payable amount: Rs. ", payable

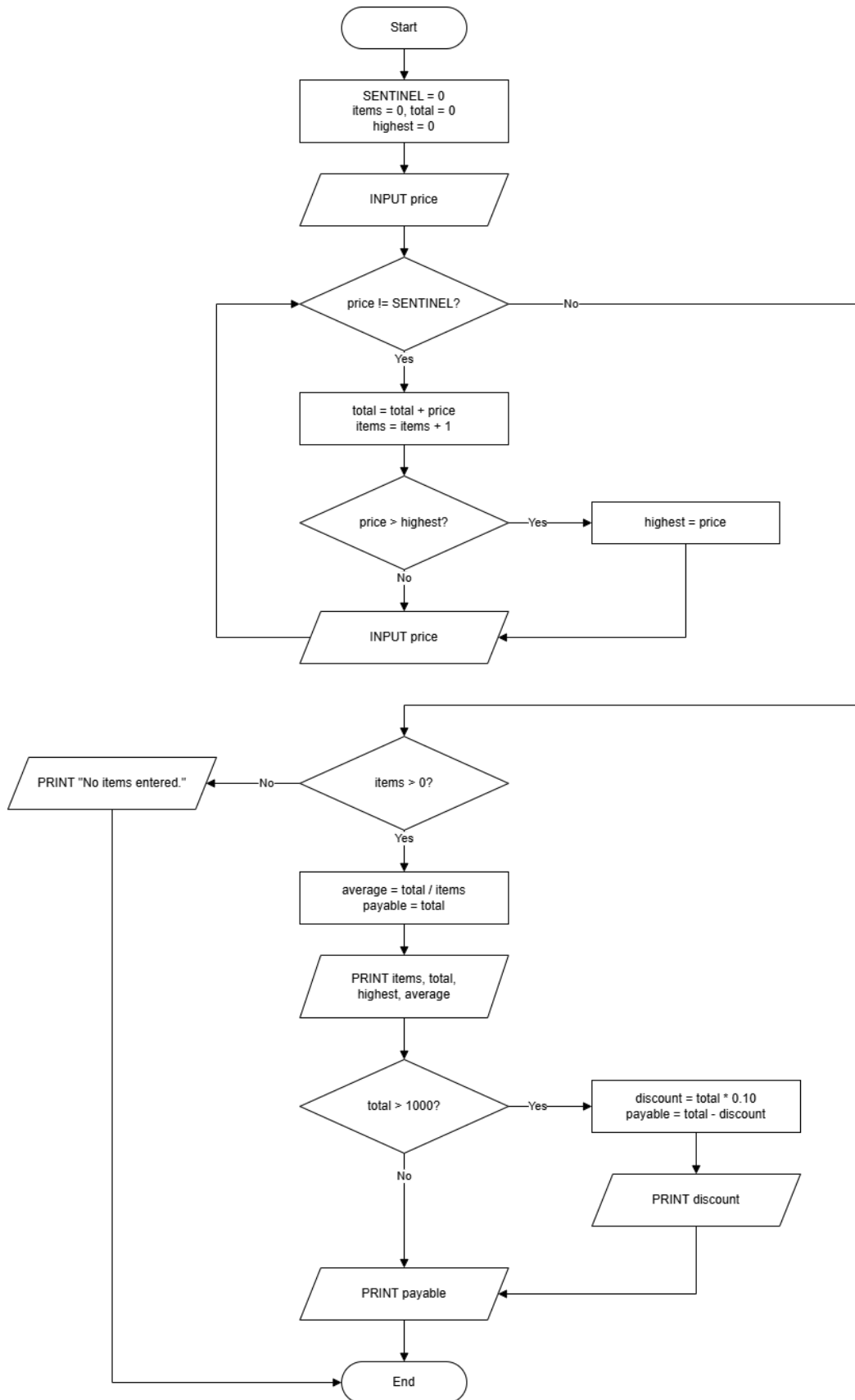
ELSE

PRINT "No items entered."

END IF

END

Flowchart:



Task 7: Using nested loops, print the following three outputs (one program or three separate programs): (the images are attached in the original lab manual).

Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5
6      // Pattern 1
7      for (int row = 1; row <= 4; row++) {
8
9          for (int col = 1; col <= row; col++) {
10             cout << row << " ";
11         }
12
13         cout << endl;
14     }
15     cout << endl;
16
17     // Pattern 2
18     for (int row = 4; row >= 1; row--) {
19
20         for (char ch = 'A'; ch < 'A' + row; ch++) {
21             cout << ch << " ";
22         }
23
24         cout << endl;
25     }
26     cout << endl;
27
28     // Pattern 3
29     for (int row = 1; row <= 5; row++) {
30
31         for (int col = 1; col <= 5; col++) {
32             cout << row * col << "\t";
33         }
34
35         cout << endl;
36     }
37
38     return 0;
39 }
40
41
42
43 }
```

Output:

```
1
2 2
3 3 3
4 4 4 4

A B C D
A B C
A B
A

1      2      3      4      5
2      4      6      8      10
3      6      9      12     15
4      8      12     16     20
5      10     15     20     25
```

Pseudocode:

START

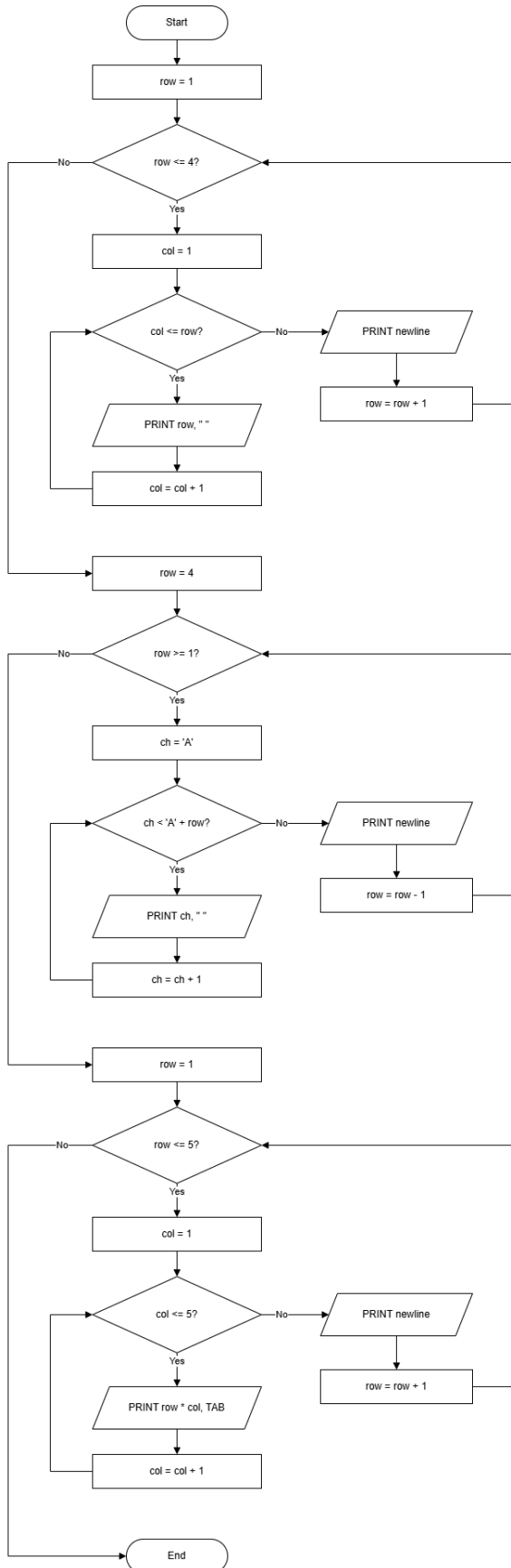
```
FOR row = 1 TO 4
  FOR col = 1 TO row
    PRINT row, " "
  END FOR
  PRINT newline
END FOR
```

```
FOR row = 4 DOWN TO 1
  FOR ch = 'A' TO 'A' + row - 1
    PRINT ch, " "
  END FOR
  PRINT newline
END FOR
```

```
FOR row = 1 TO 5
  FOR col = 1 TO 5
    PRINT row * col, TAB
  END FOR
  PRINT newline
END FOR
```

END

Flowchart:



Task 8: Assume that you want to generate a table of multiples of any given number. Write a program that allows the user to enter the number and then generates the table, formatting it into 10 columns and 5 lines.

Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int num;
6
7      cout << "Enter a number: ";
8      cin >> num;
9
10     for (int i = 1; i <= 50; i++) {
11
12         cout << num * i << "\t";
13
14         if (i % 10 == 0)
15             cout << endl;
16
17     }
18
19     return 0;
20 }
21
```

Output:

```
Enter a number: 8
8      16      24      32      40      48      56      64      72      80
88      96      104     112     120     128     136     144     152     160
168     176     184     192     200     208     216     224     232     240
248     256     264     272     280     288     296     304     312     320
328     336     344     352     360     368     376     384     392     400
```

Pseudocode:

START

INPUT num

FOR i = 1 TO 50

PRINT num * i, TAB

IF i MOD 10 == 0 THEN

PRINT newline

END IF

END FOR

END

Flowchart:

